Reverse Engineering: Toy Helicopter

Cecilia Martinez Rodriguez

4/20/2026

Design Intent:

The primary design intent for this project was to reverse engineer a toy helicopter assembly. We utilized a top-down approach, where assembly-level layouts drove part-level geometry to ensure proper fitting across individual components (top of body, bottom of body, cockpit, landing gear, and two propellers). As most toys are manufactured in this way, the geometry was modeled with injection molding in mind. Small clearance offsets were incorporated for shafts in the propeller hubs, though approximations with basic measurements were also incorporated throughout. By establishing master geometry, both for the initial body split and the cockpit interior volume, we ensured that changes to the core cabin dimensions propagate correctly to the mating parts.

Simplifications & Assumptions:

Complex internal aesthetic textures and minor non-functional surface details were simplified or omitted to focus on structural accuracy and fit. On occasion, mechanical mating parts were simplified to avoid repetition.

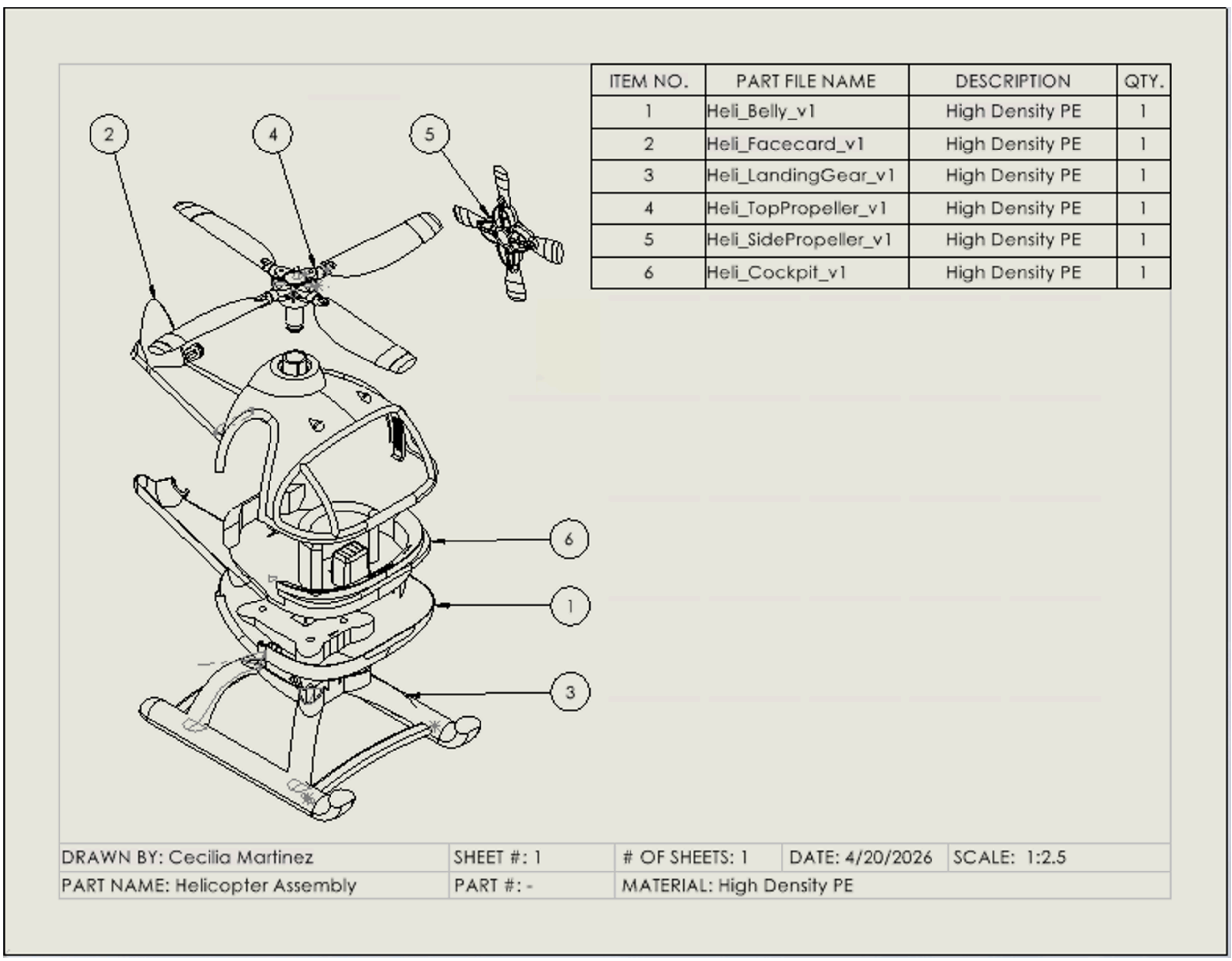
Modeling Difficulties:

Modeling this toy presented several technical challenges that required iterating the workflow. I initially faced issues when using the Combine/Subtract tool due to SolidWorks' restrictions on editing geometry in-context across different files.

Moreover, I faced challenges with my sequence in the modeling plan, as shelling the cabin too early caused me to have to rearrange my feature manager and start some features anew, especially with the top rotor hub and the landing gear mate.

Material: GreenToys are made from recycled plastic, therefore the material chosen for the modeled toy parts was HDPE, one of the most common recycled plastics (used in detergent bottles), especially for sturdy objects like toys.

Assembly Drawing:



ITEM NO.	PART FILE NAME	DESCRIPTION	QTY.
1	Heli_Belly_v1	High Density PE	1
2	Heli_Facecard_v1	High Density PE	1
3	Heli_LandingGear_v1	High Density PE	1
4	Heli_TopPropeller_v1	High Density PE	1
5	Heli_SidePropeller_v1	High Density PE	1
6	Heli_Cockpit_v1	High Density PE	1

Real Helicopter Toy Image (MACY's)

References

Molly. (2025, June 17). *Guide to plastics in toys industry: Types, safety, and Sustainability*. Weijun Toys. https://www.weijuntoy.com/plastics-in-toys-the-ultimate-guide/

File Index:

Part Files

Heli_Facecard_v1

Heli_Belly_v1

Heli_SidePropeller_v1

Heli_TopPropeller_v1

Heli_Cockpit_v1

Assembly File

Heli_Assembly_v1

All other files were auxiliary.